

Instructions

Allowed tools: Paper, pen, five pages of notes and dictionary.

Write your name and personal number here:

.....

Write the secret number given to you on EVERY page and here:

.....

If you want your result sent to you by email, write your email address here:

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This is the exam for this course, and it will contribute to the final grade of 50%; 30% of the final grade will be based on the project, 10% on the lab queries, and 10% on discussion/pre-lecture queries. To pass the course, you need to pass all labs with at least 50% and get at least 40% on this exam.

Minimum score per letter grade (truncated, not rounded)

- A: 92%
- B: 84%
- C: 76%
- D: 68%
- E: 60%
- Fx: 55%
- F: <55%

Each question should be answered within the provided space. If you need more space, write it on the back of the paper.

You can answer in either English or Swedish.

Arne can be reached at 070-6951045 for questions and will call in a few times during the exam. Petras and Samuel will come by to take questions.

Formulas that you might find helpful (from wikipedia):

$$H = - \sum_i p_i \log_2(p_i)$$

$$I(X; Y) = \sum_{y \in Y} \sum_{x \in X} p(x, y) \log \frac{p(x, y)}{p(x) p(y)},$$

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)}$$

$$S_{ij} = \frac{1}{\lambda} \log \frac{p_{ij}}{q_i q_j}$$

In computing S_{ij} , we will take $\lambda=1$.

q_i is the marginal probability of occurrence of amino acid of type i .

p_{ij} is the joint probability of the pair of amino acid types i and j .

$\log$ is base-2.

Question 1: Accessing information in sequence databases (10 p):

- Describe pros and cons using command-line and Web-based (through ordinary and genomic browsers) access to the databases,
- Analyze pros and cons of using accession numbers and standard names of genes and proteins,
- Describe building efficient queries.

Question 2: Pairwise sequence alignment (10 p):

- A. Describe the Needleman-Wunsch algorithm,
- B. Describe the Smith-Waterman algorithm,
- C. Analyze similarities and differences between those two algorithms.

Question 3: BLAST search algorithm (10 p):

- A. Describe the seeding stage,
- B. Describe the extension stage,
- C. Describe the evaluation stage.

Question 4 (10p) Ekman Ch 4.

Consider the following network to classify 28x28 pixel images of handwritten digits, into classes 0-9.

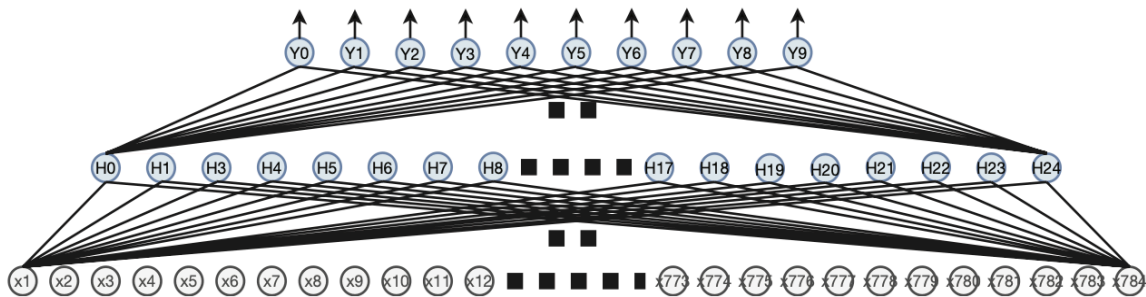


Figure 4-3 Network for digit classification. A large number of neurons and connections have been omitted from the figure to make it less cluttered. In reality, each neuron in a layer is connected to all the neurons in the next layer.

Now your goal is to accomplish the same task, but using 2D convolutional layer or layers rather than the densely connected layer shown. Assume the output layer will continue to be 10 neurons, densely connected to the layer below it.

2 pts: What needs to happen to the input layer, compared to that shown? Be explicit and provide numbers in your answer.

Propose a convolutional replacement for the 24-neuron densely connected hidden layer shown. There are many possible answers, you will be graded on numerical consistency and to some extent feasibility. You may provide text justification, if you think your choices are good but unconventional.

2 pts: What are the width and (if applicable) height dimensions?

2 pts: What is the stride?

2 pts: What is the channel depth?

Now consider the possibility of adding one hidden layer, above and in addition to the one shown.

2 pts: Is this necessary? Argue your point in the context of your other answers above.

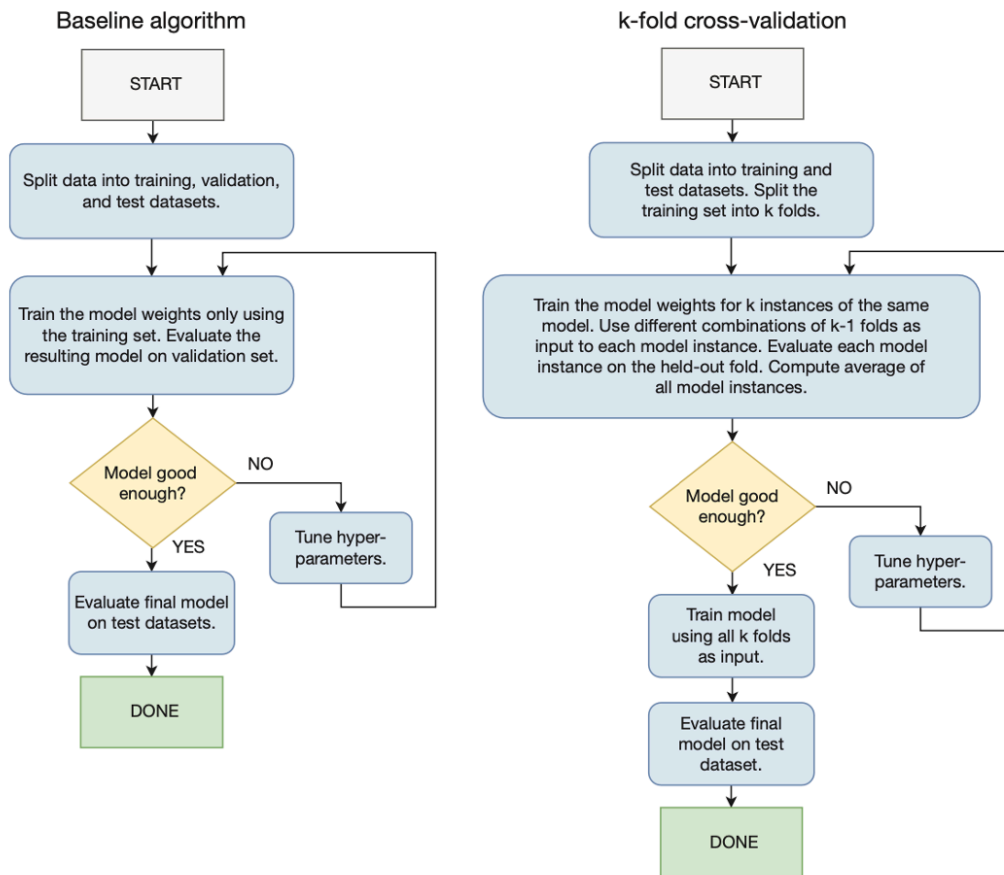
Question 5 (Ch 5, k-fold cross validation)

Figure 5-14 Tuning hyperparameters with a validation dataset (left) and using k-fold cross-validation (right)

Consider k-fold cross-validation compared to the baseline of having test, validation, and train. Which one implies more training time? How much more? Be quantitative, and justify. You are allowed to state only a minimum or only a maximum, if justified.

What is the advantage of k-fold cross-validation, compared to baseline?

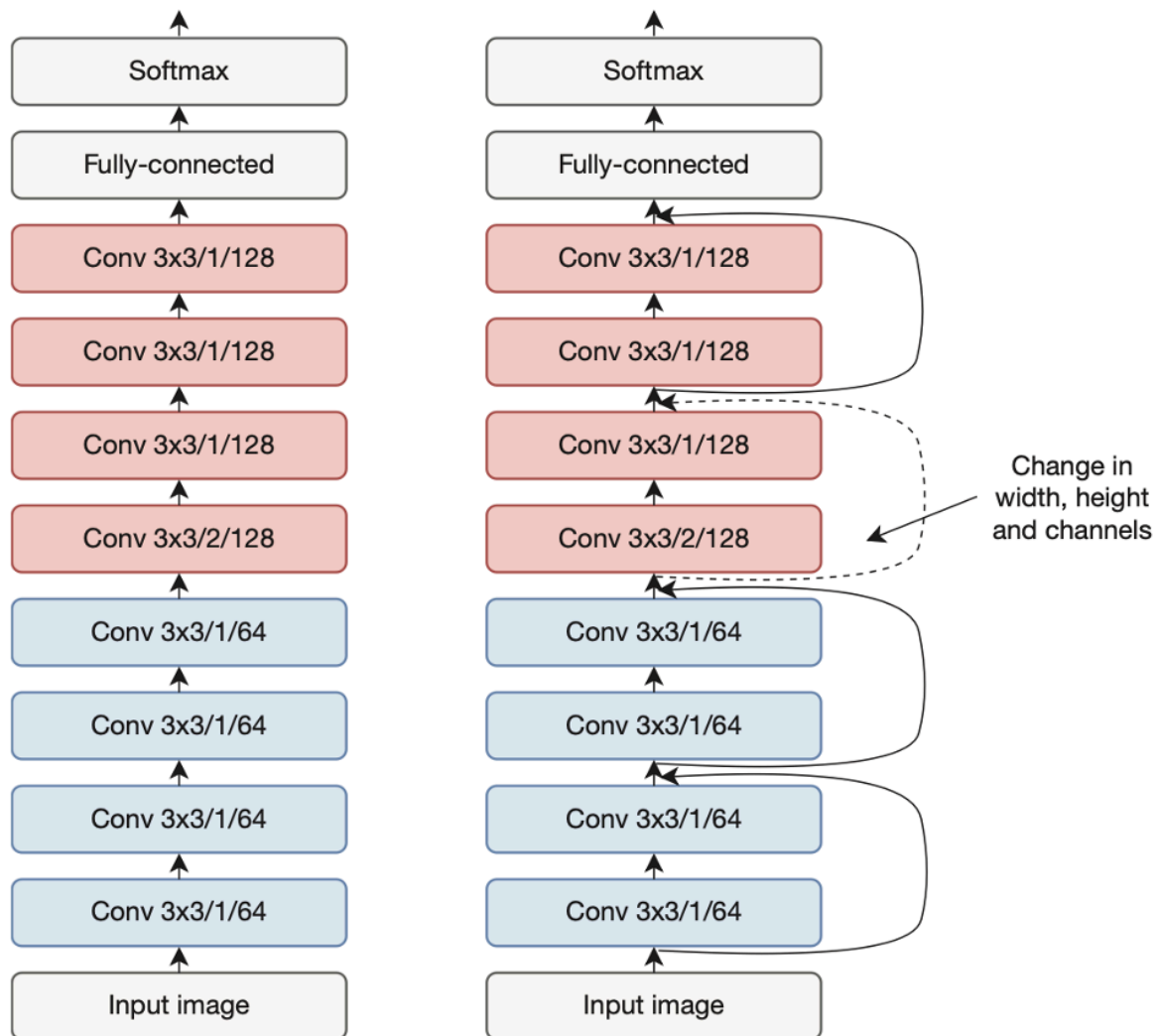
Question 6 (Ch 8, skip connections)

Figure 8-5 Left: Baseline network without skip connections. Right: ResNet-style network with skip connections. The dashed skip connection indicates that the input and output dimensions of the block do not match (details are discussed later in the chapter). The figure is simplified in that it does not explicitly show the activation functions being applied after the skip connections.

Now think about what happens to a deep neural network which has skip connections, as opposed to one without them. Give the justification for skip connections in deep networks. What problem do they solve? What is the theory for why it should work?

Note the skip connection spanning layers with a change of dimensions. Can a skip connection be made in this case? How can this skip connection be made to work? Make a sketch and provide the type and parameters of any new or modified neural layer.

Question 7 (Transformers)

A) How does the attention mechanism in Transformer models improve protein function prediction compared to traditional sequence alignment methods like MSA? (5p)

B) What are the main challenges of applying pre-trained language models like ESM or PROT-T5 to proteins with low evolutionary conservation, and how can these be addressed (5p)?

Question 8:

A) What innovations in AlphaFold2's architecture contributed to its significant improvement over earlier structure prediction methods? (5p)

B) Despite AlphaFold2's success, what are its known limitations when applied to challenging protein targets such as disordered regions, multi-domain assemblies, or membrane proteins? (5p)